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Multi-UAV cooperative task scheduling and trajectory optimization system under communication constraints

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ABSTRACT

In recent years, Unmanned Aerial Vehicles (UAVs) have been widely used in both military and civilian applications. Due to the complexity of dynamic environments and task types, it is often difficult for a single UAV to meet the actual task requirements, which drives the development of multi-UAV cooperative task scheduling and trajectory optimization techniques. However, in actual scenarios, when UAVs collaborate to execute tasks, they are subject to various constraints, including UAV performance constraints, task condition constraints, and communication constraints. This paper considers the task scheduling of multi-UAV under the condition of limited energy at communication base stations. This type of optimization problem generally belongs to the NP-hard problem category, and Reinforcement Learning (RL) algorithms have significant advantages in solving this type of problem. Therefore, this paper proposes a Temporal Task Scheduling (TTS) algorithm based on the classic Multi-Agent Reinforcement Learning (MARL) algorithm framework. The network architecture of TTS algorithm incorporates Multi-Layer Perception (MLP) and Gate Recurrent Unit (GRU), enhancing the fitting capability of the policy function and the memory capability of historical trajectories. In addition, an entropy loss function was introduced to enhance agents' abilities to explore the environment. Simulation results show that TTS algorithm has excellent convergence performance and stability, and can effectively solve the problem of multi-UAV cooperative task scheduling and trajectory optimization under communication constraints.

1. Introduction

With the rapid development of UAV technology, task scheduling and trajectory optimization of UAVs are becoming a popular research topic. Currently, there are still several complex challenges in this research area, involving task optimization, dynamic environment adaptability, and UAV obstacle avoidance capability, etc[1]. When UAVs are in a dynamic environment, rapidly changing meteorological conditions and sudden obstacles require UAVs to instantly respond and adjust their flight trajectories, which is a considerable challenge for traditional path planning algorithms[2]. Complex traffic management schemes are required when multiple UAVs are performing tasks in concert, such as avoiding collision events and optimizing overall efficiency. When UAVs are executing tasks in a coordinated manner, there is also a need to avoid collisions and optimize overall efficiency, therefore complex traffic management schemes are to be employed[3]. To deal with such problems, researchers have explored a variety of solutions[4–6], such as the Artificial Intelligence (AI) based adaptive path planning systems, and the use of machine learning algorithms to train UAVs making optimal decisions in complex environments. In addition, optimization techniques such as

Tabu Search Algorithm (TSA) and Simulated Annealing Algorithm (SAA) are integrated, and then efficient path planning schemes can be generated and the specified work is guaranteed to be completed within the time limit[7,8]. Using a combination of these strategies, UAVs can more reasonably cope with uncertain scenarios in flight.

As UAV technology continues to expand and extend its applications in multiple fields, traditional task scheduling and trajectory optimization algorithms are unable to meet a series of needs of large-scale surveillance, disaster assessment and complex environment inspection, which has prompted researchers to focus on optimization methods based on Reinforcement Learning (RL) algorithms[9,10]. In recent years, RL algorithms are outstanding due to their performance in complex environments and dynamic changes, they are gradually becoming the core of research in UAV task scheduling and trajectory optimization[11]. When faced with a large number and multiple types of tasks, a single UAV is generally difficult to meet the demands of complex tasks, which in turn requires the use of multiple UAVs to collaborate to accomplish tasks[12]. Multi-UAV pose challenges such as coordination, cooperation, and communication constraints, which are generally difficult to be handled by traditional trajectory optimization methods[13–15]. This

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has led to the research of Multi-Agent Reinforcement Learning (MARL) algorithms, which allow individual UAVs to learn environmental information to dynamically adjust their strategies and flight paths, so that they can efficiently cooperate in complex and changing task environments. Given that MARL methods can continuously optimize strategies through simulation without the need to manually define decision-making methods, they significantly improve adaptability and robustness of systems[16]. The technical support given by the current rapid development of AI, as well as factors such as the increasing demand for UAV formations operating efficiently and autonomously, they promote the application and research of MARL algorithms in multi-UAV cooperative task scheduling together[17]. The ultimate goal is to explore trajectory optimization methods that can solve complex tasks and guarantee high efficiency and reliability.

In terms of current technological advances as well as application requirements, it is significant to study RL-based UAV trajectory optimization methods. The MARL system uses simulation training to help UAVs reach efficient collaboration in uncertain and dynamically changing environments, which not only improves the task completion rate and accuracy, but also dramatically reduces resource consumption[18,19]. The MARL approach can effectively deal with the complex interaction of UAVs with each other, task scheduling decisions, and trajectory optimization problems, and thus improve the overall system performance while ensuring the safety of their respective maneuvers and maintaining the coherence of task objectives[20,21]. RL algorithms drive the cutting-edge research of AI in the fields of machine learning, automatic control, and multi-agent collaboration, and provide a highly scalable path planning framework with strong adaptive capability[22–24]. Based on the comparative analysis of the aforementioned literature, current research on RL algorithms holds significant importance in the field of UAV task scheduling. RL algorithms provide a highly efficient and accurate method for UAV task scheduling. However, when dynamic environments become increasingly complex, traditional RL algorithms often struggle to address the challenges at hand, this presents a challenge for RL algorithm research. Addressing the issues of slow convergence and high computational complexity during training in complex environments, developing an efficient solution algorithm is now an urgent priority.

1.1. Related work

Relevant methods for multi-UAV cooperative task scheduling and trajectory optimization have been investigated in [25–30]. Reference [25] addresses the issue that the current network architecture cannot provide sufficient fault tolerance space in real time when UAVs encounter fault nodes during task scheduling. It designs a flexible network architecture that supports dynamic fault-tolerant task scheduling for UAVs. It also proposes a Flexible network architecture Dynamic Scheduling Algorithm (FDSDA) to solve the task scheduling problem, significantly enhancing the fault tolerance capability of multi-UAV collaborative task scheduling. Reference [26] proposes a Mobile Edge Computing (MEC) model based on multi-UAV, which improves data processing efficiency by utilizing an integrated UAV network. A two-layer optimization method is established, with the upper layer focusing on optimizing UAV task scheduling and the lower layer addressing UAV flight path issues. This method treats UAVs as computing servers, which enable to optimize task scheduling and computational resource allocation strategies among UAVs under various task and service request scenarios. It also reduces latency, enhances data transmission rates, and optimizes overall network performance. Reference [27] addresses the current issues of high energy consumption and insufficient computing resources in mobile devices by developing a UAV-based aerial MEC network architecture. This architecture enables communication and task offloading services for mobile devices in areas without infrastructure, using UAVs as access points. Additionally, it proposes an energy-efficient task scheduling framework that efficiently manages and maximizes UAV en-

ergy efficiency while maintaining the computational performance required for task processing. Reference [28] addresses the challenges of trajectory optimization and network configuration in complex multi-UAV network environments, proposing an innovative multi-UAV collaborative trajectory optimization method. This method utilizes heuristic algorithms and Deep Reinforcement Learning (DRL) algorithms to optimize the flight trajectories of UAVs. It also combines Bayesian Optimization (BO) methods to estimate UAV decisions based on historical trajectory points, thereby avoiding inefficient environmental exploration, improving the convergence speed of model training, and enhancing network performance. Reference [29] investigates the problem of fair three-dimensional multi-UAV trajectory optimization in MEC systems, and a three-dimensional dynamic multi-UAV-assisted MEC system has been designed. The literature models flight trajectories of UAVs as sequences of position coordinates, and uses Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm to solve this problem, ultimately addressing the high energy consumption issues of UAVs in communication, computation, and flight. Reference [30] addresses the challenges of complex environments, and diverse requirements in current multi-UAV trajectory optimization problems. It proposes a multi-objective optimization-based collaborative trajectory planning model for multi-UAVs. This model aims to optimize UAV flight trajectories, flight times, and collaborative costs between UAVs. The third-generation Non-dominated Sorting Genetic Algorithm (NSGA-III) is employed for solution, and a Segmented Crossover Strategy (SCS) is introduced to enhance the model's computational efficiency and accelerate the algorithm's convergence speed, thereby addressing diverse practical requirements.

The aforementioned research has made significant contributions to multi-UAV task scheduling and trajectory optimization under various networking and computing settings. Different from many existing studies that primarily impose a communication-range constraint or optimize trajectories/scheduling without explicitly modeling heterogeneous communication availability, this paper focuses on a BS-assisted task scheduling and trajectory planning framework under explicit communication constraints. Specifically, we consider a coverage-limited BS (representing a limited transmit-power/energy budget), where UAVs are divided into in-coverage and out-of-coverage groups with different information availability, and out-of-coverage UAVs must perform spectrum sensing and opportunistic channel access. Meanwhile, we incorporate dynamic task arrivals with time windows and type-matching requirements, together with hard flight constraints. Based on this problem setting, we develop a MARL-based solution (TTS) with an MLP-GRU policy network to better capture static mappings and temporal dependencies, and employ a constraint-aware sampling mechanism to improve learning stability under multiple simultaneous hard constraints.

1.2. Paper contributions

The main contributions of this paper are as follows:

- **Multi-UAV Cooperative Task Scheduling and Trajectory Optimization Model:** Under conditions where UAV communication is restricted and the location and time of tasks are random, and takes into account the performance constraints of UAVs, the temporal constraints of tasks, the time constraints of tasks, the wireless communication constraints, and the spectrum sensing constraints, a multi-UAV cooperative task scheduling and trajectory optimization model under communication constraints is established, and the optimization objectives are modeled.
- **Optimization Algorithm Design:** Based on the MARL framework, we propose the TTS algorithm for the formulated problem. We adopt PPO-style clipped policy optimization with GAE and entropy regularization as standard training techniques to improve stability and exploration. The key algorithmic ingredient is a constraint-aware on-policy data collection mechanism (rollback and record

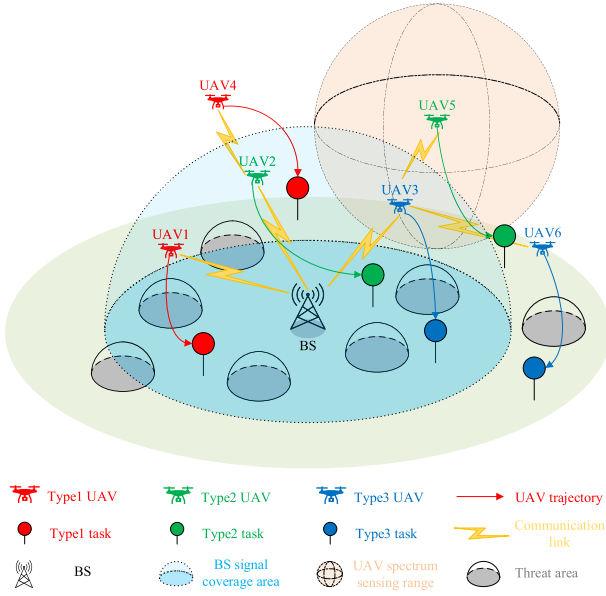


Fig. 1. Multi-UAV cooperative task scheduling and trajectory optimization system model under communication constraints.

cancellation) to prevent infeasible samples from entering policy updates under multiple hard constraints.

- **Network Design:** By training all UAV policy networks together, the coordination between UAVs is enhanced, enabling multi-UAV to collaborate on tasks and reducing conflicts between them. Additionally, MLP and GRU are introduced on top of the Actor network and Critic network, significantly enhancing networks' ability to fit the strategy function and its memory of UAVs' exploration history.

The rest of this paper is organized as follows. Section II establishes the model used in this paper and models the optimization problem. Section III provides a detailed introduction to the optimization algorithm proposed in this paper, and describes the learning system designed in this paper. Section IV presents the relevant simulation results and conducts a comparative analysis. Section V summarizes the entire study and draws conclusions.

2. System model and problem formulation

In this section, we model multi-UAV task scheduling and trajectory optimization under communication constraints, and the optimization problem is modeled.

2.1. System model

We consider a multi-UAV cooperative task scheduling and trajectory optimization system under communication constraints as shown in Fig. 1. We consider a situation where a cluster of UAVs is controlled for task scheduling through a communication base station (BS), but with the limited energy of the BS, it is not capable of scheduling all UAVs. It is assumed that UAVs located within the signal coverage of the BS can communicate directly with the BS, and they have all the information about UAVs held by the BS. As for UAVs located outside the signal coverage of the BS, they need to use spectrum sensing technology to access the communication channel on their own for communication.

In this model, there exists a BS and U UAVs, and the set of UAVs is $\mathcal{U} = \{1, 2, \dots, M\}$. Considering the limited energy of the BS, its signal coverage is set to R^B . The BS energy limitation is modeled using a coverage-limited proxy: the BS is assumed to operate under a limited transmit-power/energy budget, which results in a finite coverage radius R^B . We do not explicitly model time-varying BS energy depletion

nor optimize BS energy usage in the current formulation; instead, different BS energy/power levels are reflected by varying R^B in simulations. Assume that all UAVs perform their tasks in T continuous time slots, where $\mathcal{T} = \{1, 2, \dots, T\}$ is the set in T continuous time slots, and the length of each time slot is $\Delta\tau$. At time slot $t (t \in \mathcal{T})$, the position coordinates of UAV $u (u \in \mathcal{U})$ are defined as $\mathbf{q}_u(t) = [x_u(t), y_u(t), z_u(t)]$. In this system, considering that the communication through relay UAVs will generate time delay, and this system has high requirements for communication efficiency, in order to ensure better communication efficiency, therefore, only the communication within one hop from the BS to the UAV is considered, i.e., there is no relay equipment. Define the set of communication channels as $\mathcal{N} = \{1, 2, \dots, N\}$, the total number of frequency band resources as N , and both the BS and each UAV need to be allocated a frequency band when communicating. When UAVs are located within the signal coverage of the BS, the BS can directly assign the corresponding frequency band for these UAVs. On the contrary, UAVs need to carry out spectrum sensing and access the remaining available frequency bands on their own for communication.

2.1.1. UAVs Model

For the sake of generality, it is assumed that the flyable area of UAVs in three-dimensional space is limited to $[X^{\min}, X^{\max}] \times [Y^{\min}, Y^{\max}] \times [Z^{\min}, Z^{\max}]$, and the BS is located at coordinates $(x^B, y^B, 0)$. Due to the flexibility of UAVs flight, they can dynamically change their positional coordinates in the three-dimensional space, and unlike models that assume UAVs are at a fixed altitude, this model is able to realistically capture different flight altitudes of UAVs.

In this model, the flyable range of UAVs is $\mathbb{U}$, which is defined as

$$\mathbb{U} = \{[x_u(t), y_u(t), z_u(t)] \in \mathbb{R}^3 \mid X^{\min} \leq x_u \leq X^{\max}, Y^{\min} \leq y_u \leq Y^{\max}, Z^{\min} \leq z_u \leq Z^{\max}, t \in \mathcal{T}\} \quad (1)$$

where $\mathbb{R}^3$ denotes the three-dimensional space.

At time slot t , we assume that there are U_t^I UAVs located within the signal coverage of the BS, where $\mathcal{U}_t^I = \{1, 2, \dots, U_t^I\}$, the number of frequency bands occupied by the BS and these UAVs is $N_t^I + 1$. Assume that there are U_t^O UAVs located outside the signal coverage of the BS, where $\mathcal{U}_t^O = \{1, 2, \dots, U_t^O\}$, the number of frequency bands occupied by these UAVs is $N - (N_t^I + 1)$. And the following conditions exist

$$\mathcal{U} = \mathcal{U}_t^I \cup \mathcal{U}_t^O \quad (2)$$

$$U = U_t^I + U_t^O \quad (3)$$

$$N - (N_t^I + 1) < U_t^O \quad (4)$$

Assume that during any time slot t , UAV $u^I (u^I \in \mathcal{U}_t^I)$ has been allocated communication band resources at the beginning of this time slot, then it can directly perform its tasks. UAV $u^O (u^O \in \mathcal{U}_t^O)$, on the other hand, needs to first perform spectrum sensing for a duration of τ , and then it executes the corresponding task after successfully accessing the communication channel for a duration of $\Delta\tau - \tau$. Since UAV u^O performs its tasks in less time than UAV u^I , in general, the flight path length of UAV u^O will not exceed that of UAV u^I , i.e.

$$l_{u^O}(t) \leq l_{u^I}(t) \quad (5)$$

where $l_{u^O}(t)$ and $l_{u^I}(t)$ denote the flight path lengths of UAVs u^O and u^I in time slot t respectively.

To ensure UAVs to fly safer during task execution, we referenced the path planning modeling approach in Reference [31], and introduced constraints for safe distances between UAVs, maximum turn angles and threat area risk distances.

In order to avoid collisions between UAVs, we define the safe distance between UAVs as d^S . At any time slot t , the distance between any two UAVs should be no less than d^S , i.e.

$$\|\mathbf{q}_u(t) - \mathbf{q}_{u'}(t)\|_2 \geq d^S \quad (6)$$

where $\|\cdot\|_2$ denotes the Euclidean paradigm number.

The UAV needs to ensure that its turning angle is not too large during flight, defining the maximum turning angle of any UAV as θ . At any time slot t , the turning angle of any UAV should be no greater than θ , i.e.

$$\frac{[\mathbf{q}_u(t+2) - \mathbf{q}_u(t+1)][\mathbf{q}_u(t+1) - \mathbf{q}_u(t)]^T}{\|\mathbf{q}_u(t+2) - \mathbf{q}_u(t+1)\|_2 \|\mathbf{q}_u(t+1) - \mathbf{q}_u(t)\|_2} \geq \cos \theta \quad (7)$$

In this model, there are J threat areas, the set of threat regions is $\mathcal{J} = \{1, 2, \dots, J\}$, and threat areas are modeled as hemispheres. For the j th ($j \in \mathcal{J}$) threat area, the coordinates of its center location are $\mathbf{h}_j = (x_j, y_j, 0)$ and its radius is R_j . UAVs also need to ensure that their position coordinates are outside of the threat area during flight, i.e.

$$\|\mathbf{q}_u(t) - \mathbf{h}_j\|_2 > R_j \quad (8)$$

2.1.2. Tasks model

Considering the complexity of UAVs executing tasks in dynamic environments, we established time slot constraints for task execution when establishing tasks model. We also differentiated task types and set constraints for UAVs to successfully complete designated tasks, enabling them to accomplish tasks more efficiently. In this model, task target points are dynamic, and their appearance location and time are random. Each target point sends its position and time information to the BS when it appears, and the BS organizes the information after receiving it, then makes a decision and assigns the UAV to perform the corresponding task.

We assume that the total number of tasks is M , and there are K types of tasks, where $\mathcal{M} = \{1, 2, \dots, M\}$ denotes the set of task numbers and $\mathcal{K} = \{1, 2, \dots, K\}$ denotes the set of task type numbers. For UAV clusters, there are likewise K types of UAVs, the set of UAV type numbers is likewise $\mathcal{K}$, and the k th ($k \in \mathcal{K}$) type of task can only be accomplished by the k th type of UAV.

Physical meaning of type matching: The UAV/task “type” represents practical heterogeneity in onboard payloads and capabilities and corresponding task requirements. Therefore, the constraint that a task of type k can only be executed by a UAV of type k models capability feasibility rather than an artificial partition. Moreover, although tasks are type-specific, the overall problem is not equivalent to solving k independent subproblems in parallel because all UAVs share the same spatio-temporal environment and communication resources, and they are jointly coupled by hard flight constraints (collision avoidance/safety distance, turning-angle limits, and threat avoidance). These couplings induce cross-type competition and coordination in space, time, and communication.

For the m th ($m \in \mathcal{M}$) task, it belongs to the k th type of task, define $[t_m^S, t_m^E]$ as the execution time window for task m and t_m as the execution time of task m , where t_m^S and t_m^E ($t_m^S, t_m^E \in \mathcal{T}$) denote the execution start time slot and execution completion time slot of task m respectively. Define $[T_m^S, T_m^E]$ as the occurrence time window for task m , where T_m^S and T_m^E ($T_m^S, T_m^E \in \mathcal{T}$) denote the appearance time slot and disappearance time slot of task m respectively. Therefore, for the time window of task m , the following constraints exist

$$T_m^S \leq t_m^S < t_m^E \leq T_m^E \quad (9)$$

Feasibility and constraint interaction: The introduced constraints may interact and can potentially lead to infeasible decisions under dense task arrivals or limited communication. In our formulation, feasibility is handled in two layers. First, the environment and task-generation settings are chosen such that feasible motions exist under nominal conditions, while tasks that cannot be served within their time windows are allowed to remain unaccomplished and are penalized in the reward. Second, during on-policy learning, if a sampled action would result in any hard constraint violation, the action is cancelled and the agent rolls back to the last feasible state, and the corresponding transition is removed from the on-policy record before re-sampling a new action (roll-back and record cancellation). This mechanism prevents infeasible samples from entering policy updates, thereby improving training stability when multiple hard constraints are imposed simultaneously.

For the execution time of task m , the following constraints exist

$$t_m \leq T_m^E - T_m^S \quad (10)$$

In this model, task points are modeled as spherical areas. The location coordinates of task m are defined as $\mathbf{w}_m = (x_m, y_m, z_m)$ and the range radius of all task points is R^M . Assume that UAV u is assigned to perform task m , when the distance between UAV u and the location of task m is less than R^M , the UAV arrives at the designated target point and completes the task after an execution time t_m . Therefore, the constraint for UAV u to reach the location of task m is defined as

$$\|\mathbf{q}_u(t) - \mathbf{w}_m\|_2 < R^M \quad (11)$$

There is a specific association between tasks and task types. We assume that the task type association matrix is $\Phi = [\phi_{m,k}]_{M \times K}$, where $\phi_{m,k}$ denotes the association coefficient between task m and task type k , which is defined as

$$\phi_{m,k} = \begin{cases} 1, & \text{if task } m \text{ belongs to task type } k \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

Define $f(m, k)$ to be a binary function, where $f(m, k) = 1$ denotes that task m was executed successfully, and $f(m, k) = 0$ denotes that task m was executed unsuccessfully, i.e.

$$f(m, k) = \begin{cases} 1, & \text{if task } m \text{ is executed successfully} \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

Based on the above equation, define the task scheduling matrix as $\mathbf{F} = [f(m, k)]_{M \times K}$.

Assume that the communication association matrix of the UAV is $\mathbf{C} = [c_{u,t}]_{U \times T}$, where $c_{u,t}$ denotes whether UAV u successfully accesses the communication channel at time slot t , which is defined as

$$c_{u,t} = \begin{cases} 1, & \text{if UAV } u \text{ accesses the communication} \\ & \text{channel at time slot } t \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

Assume that the task association matrix of the UAV is $\mathbf{G} = [g_{u,m}]_{U \times M}$, where $g_{u,m}$ denotes whether UAV u participates in task m , which is defined as

$$g_{u,m} = \begin{cases} 1, & \text{if UAV } u \text{ participates in task } m \text{ and } u \\ & \text{is of the same type as } m \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

Define the function $t(u, m)$ to denote the time consumed by UAV u in task m , i.e.

$$t(u, m) = \begin{cases} t_m, & \text{if } g_{u,m} = 1 \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

2.1.3. Wireless communication model

In most modern wireless communication networks, co-channel interference is a primary factor affecting communication quality. To ensure seamless communication for UAVs, we designed a Signal-to-Interference-plus-Noise Ratio (SINR) model, using SINR as the evaluation metric for UAVs' wireless communication. In this model, it is considered that UAVs will be affected by external interference during the communication process, the SINR at the receiving end of the signal needs to be calculated. When communication is performed, it is categorized into the following scenarios:

Scenario 1: The BS communicates with UAV u^I within the signal coverage of the BS, where the communication channel is a Line of Sight (LOS) channel and the UAV u^I is subject to Gaussian noise interference.

Scenario 2: The UAV u^I communicates with the UAV u^O outside the signal coverage of the BS, where the communication channel is a LOS channel and the UAV u^O is subject to Gaussian noise interference.

Scenario 3: The UAV u^O communicates with the UAV u^I . The communication channel is a LOS channel and the UAV u^I is subject to co-channel interference from other UAVs as well as Gaussian noise.

Scenario 4: The UAV u^I communicates with the BS, the communication channel is a probabilistic LOS channel and the BS is subject to Gaussian noise interference.

Assume that the transmitting end of the signal is p , and the receiving end of the signal is q . In scenarios 1 and 2, the SINR of the received signal is defined as

$$SINR(q) = \frac{Ph_{p,q}d_{p,q}^{-\alpha}}{\sigma_n^2} \quad (17)$$

where P denotes the signal transmission power, $P = P_B$ in scenario 1 and $P = P_U$ in scenario 2, P_B and P_U denote the signal transmission power of the BS and the UAV respectively. $h_{p,q}$ denotes the channel power gain between p and q , $d_{p,q}$ denotes the spatial distance between p and q , α denotes the path loss exponent, and σ_n^2 denotes Gaussian noise power.

In scenario 3, the SINR of the received signal is defined as

$$SINR(q) = \frac{P_U h_{p,q} d_{p,q}^{-\alpha}}{I_q + \sigma_n^2} \quad (18)$$

where I_q denotes the total co-channel interference power received by the signal receiver q , i.e., the total co-channel interference power received by the UAV u^I , which is defined as

$$I_q = \sum_{p' \in U, p' \neq p} I_{p',q} \quad (19)$$

where $I_{p',q}$ denotes the co-channel interference power of the signal receiver q by the other signal transmitter p' , which is defined as

$$I_{p',q} = \begin{cases} \frac{P_U h_{p',q} d_{p',q}^{-\alpha}}{d_{p',q}^{\alpha}}, & \text{if } p' \text{ and } q \text{ are connected to the} \\ & \text{same communication channel} \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

In scenario 4, the SINR of the received signal is defined as

$$SINR(q) = \frac{P_U L(p,q) h_{p,q} d_{p,q}^{-\alpha}}{\sigma_n^2} \quad (21)$$

where $L(p,q)$ is defined as

$$L(p,q) = P_{LOS}(p,q) + \eta P_{NLOS}(p,q) \quad (22)$$

where $P_{LOS}(p,q)$ and $P_{NLOS}(p,q)$ denote the probability of occurrence of line-of-sight (LOS) channels and non-line-of-sight (NLOS) channels between the signal transmitter p and the signal receiver q respectively, and $P_{LOS}(p,q) + P_{NLOS}(p,q) = 1$. η denotes the overfading factor of the NLOS channel, and $\eta < 1$.

$P_{LOS}(p,q)$ is defined as

$$P_{LOS}(p,q) = \frac{1}{1 + e^{-B(\beta-C)}} \quad (23)$$

where B and C are constants whose values depend on the actual environment, and β is the angle between the direction of the signal transmitter p to the signal receiver q and the horizontal plane.

In addition, when UAV u^I communicates with UAV u^O , UAV u^I must be located within the communication range of UAV u^O , i.e.

$$\|\mathbf{q}_{u^I}(t) - \mathbf{q}_{u^O}(t)\|_2 \leq R_{u^O}^C \quad (24)$$

where $R_{u^O}^C$ denotes the communication range radius of the UAV u^O .

For any UAV $u(u \in U)$, the radius of its communication range is always not larger than the radius of the signal coverage of the BS, i.e.

$$R_u^C \leq R^B \quad (25)$$

where R_u^C denotes the radius of the communication range of the UAV u .

2.1.4. Spectrum sensing model

We assume that at time slot t , any UAV u^I has all the information of the BS at this time slot. Therefore, the UAV u^O only needs to communicate with the UAV u^I , thereby performing spatial spectrum sensing and accessing the remaining available communication bands on its own. Therefore, this spectrum sensing model is consistent with the wireless communication model of Scenario 2 above.

The total energy of the signal received by the UAV u^O is defined as

$$\Gamma_{u^O} = \frac{1}{N_s} \sum_{n=0}^{N_s-1} \left| \sqrt{\frac{P_U h_{u^I,u^O}}{d_{u^I,u^O}^{\alpha_{AA}}}} s_{u^I}(n) + n_0(n) \right|^2 \quad (26)$$

where N_s is the total number of samples, n is the sampling index, α_{AA} is the path loss exponent from one air node to another air node, $s_{u^I}(n)$ denotes the n th sampled signal of signal transmitter u^I , and $n_0(n)$ denotes co-channel interference signals from other UAVs that have accessed the communication channel.

In this model, when the total energy of the signals received by the UAV is greater than a predetermined energy value, the UAV is considered to have successfully sensed the current spectrum and accessed the corresponding communication channel. Conversely, the UAV does not perceive the current spectrum. Therefore, the condition for the UAV u^O to successfully perform spectrum sensing is defined as

$$\Gamma_{u^O} \geq \Gamma_0 \quad (27)$$

where Γ_0 denotes the spectrum perception value threshold of the UAV, i.e., the minimum energy threshold for the UAV to successfully perceive the spectrum.

The above method provides an efficient spectrum sensing strategy and balances spectrum sharing and co-channel interference between UAVs.

2.2. Problem formulation

Our goal is to rationally schedule tasks to be performed and optimize flight trajectories of all UAVs in the presence of communication constraints of UAVs, thus maximizing the comprehensive performance of the system. In this model, three main aspects of optimization are considered, which include task performance, communication performance and consumption time. Thus, the objective function in this model can be defined as

$$\begin{aligned} \text{P1 : } \max_{\mathbf{F}, \mathbf{C}, \mathbf{G}} & \left[\lambda_1 \sum_{m=1}^M \sum_{k=1}^K f(m,k) + \lambda_2 \sum_{t=1}^T \sum_{u=1}^U c_{u,t} \right. \\ & \left. - \lambda_3 \sum_{m=1}^M \sum_{u=1}^U t(u,m) \right] \\ \text{s.t. } & \text{C1 : (1) - (8)} \\ & \text{C2 : (9), (10), (11)} \\ & \text{C3 : (24), (25)} \\ & \text{C4 : (27)} \end{aligned} \quad (28)$$

where λ_1 , λ_2 and λ_3 are all weighting parameters, and $\lambda_1 + \lambda_2 + \lambda_3 = 1$. By introducing weight parameters, the priority of sub-objectives within the objective function can be dynamically adjusted. When actual application scenarios change, the importance of sub-objectives may also shift. Simply modifying the weight parameters corresponding to each sub-objective alters its priority without requiring changes to the sub-objective functions themselves. Consequently, this weight parameter approach significantly simplifies the algorithm's computational complexity. Additionally, employing a linear weighting approach to derive the final objective function significantly enhances gradient stability while reducing the risk of numerical divergence during training. $\mathbf{F}$, $\mathbf{C}$ and $\mathbf{G}$ denote the feasible solution space for the three optimization variables of task scheduling matrix, communication association matrix and task association matrix respectively. C1 is the UAV constraint group, C2 is the

task constraint group, C3 is the communication constraint group, and C4 is the spectrum sensing constraint. Since problem P1 involves nonconvex constraints (6), (7) and (8), P1 is an NP-hard problem. Considering that traditional swarm-based optimization algorithms, such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA), exhibit low efficiency when solving NP-hard problems and poor generalization capabilities in dynamically changing environments, this paper proposes TTS algorithm based on MARL framework to address Problem P1. The advantage of TTS algorithm lies in its ability to make real-time decisions in dynamically changing environments. It features high training efficiency and rapid response speeds, effectively enhancing coordination among UAVs when addressing the multi-UAV cooperative task scheduling and trajectory optimization problem proposed in this paper.

3. Algorithm design and analysis

In this section, we give the corresponding solution to Problem P1, propose TTS algorithm, introduce the components and specific structure of TTS algorithm in detail, and design a MARL decision model for multi-UAVs to collaborate on task scheduling in three-dimensional space. Finally, we provide a specific analysis for the complexity of TTS algorithm used in this subsection.

3.1. TTS algorithm design

In order to solve the optimization problem mentioned above, we design a multi-UAV cooperative MARL decision-making method to optimize the task scheduling of UAVs.

In our model, each agent makes its own decisions and plans its flight trajectory separately. It should be noted that UAVs within the signal coverage of the BS directly communicate with the BS and obtain global information, while UAVs outside the signal coverage of the BS need to perform spectrum sensing and communicate with UAVs within range to obtain global information. We define the decision model of any agent as a quintuple $(S, \mathcal{O}, \mathcal{A}, \mathcal{R}, \mathcal{P}_r)$, each element of this quintuple is defined as follows

3.1.1. Action space $\mathcal{A}$

According to Problem P1, the flight constraints of UAVs include the safe distance, the turning angle and the distance between UAVs and threat areas. Therefore, in order to simplify the arithmetic process, the action of UAV $u (u \in \mathcal{U})$ at time slot $t (t \in \mathcal{T})$ is defined as

$$a_{u,t} = \{v_u^X(t), v_u^Y(t), v_u^Z(t), \rho_u(t)\} \quad (29)$$

where $[v_u^X(t), v_u^Y(t), v_u^Z(t)]$ denotes the flight velocity vector of UAV u at time slot t , $\rho_u(t)$ denotes the communication channel accessed by UAV u at time slot t , and $\rho_u(t) \in \mathcal{N}$.

Furthermore, consider that the action space contains both continuous and discrete variables, therefore the discrete variable $\rho_u(t)$ is made continuous. Assumption $\rho_u(t) \in [0, 1]$, which is used to adjust the communication channel index $\rho_u(t)$. After the transformation, the UAV u will access the $[\rho_u(t) \cdot N]$ th channel for communication at time slot t , where $[\cdot]$ is a rounding function. At the same time, the flight velocity vector in the action space is normalized. Therefore, the action of UAV u at time slot t can be rewritten as

$$a_{u,t} = \{\bar{v}_u^X(t), \bar{v}_u^Y(t), \bar{v}_u^Z(t), \rho_u(t)\} \quad (30)$$

where $[\bar{v}_u^X(t), \bar{v}_u^Y(t), \bar{v}_u^Z(t)]$ denotes the normalized UAV flight velocity vector, which is defined as

$$\bar{\mathbf{v}}_u(t) = \left[\frac{v_u^X(t)}{(v^X)^{\text{Max}}}, \frac{v_u^Y(t)}{(v^Y)^{\text{Max}}}, \frac{v_u^Z(t)}{(v^Z)^{\text{Max}}} \right] \quad (31)$$

where $\bar{\mathbf{v}}_u(t) = [\bar{v}_u^X(t), \bar{v}_u^Y(t), \bar{v}_u^Z(t)]$, $(v^X)^{\text{Max}}$, $(v^Y)^{\text{Max}}$ and $(v^Z)^{\text{Max}}$ denote the maximum value of the velocity component of UAVs on X, Y and Z axes respectively.

Thus, the action space $\mathcal{A}$ is defined as

$$\mathcal{A} = \{a_{u,t} | u \in \mathcal{U}, t \in \mathcal{T}\} \quad (32)$$

3.1.2. Observation space $\mathcal{O}$

Each UAV is capable of observing the environment and thus obtaining partial environmental observations. The observation data consists of coordinates of the UAV's position at current time slot, the communication channel to which the UAV is connected, the spatial distance of the UAV from the target point, and the spatial distance of the UAV from the closest threat area. Therefore, the observation data of UAV u at time slot t is defined as

$$o_{u,t} = \{x_u(t), y_u(t), z_u(t), \rho_u(t), d_{u,m}(t), d_{u,j}(t)\} \quad (33)$$

where $[x_u(t), y_u(t), z_u(t)]$ denotes the positional coordinates of UAV u at time slot t . $d_{u,m}(t) = \|\mathbf{q}_u(t) - \mathbf{w}_m\|_2$, which denotes the spatial distance between UAV u and its target point $m (m \in \mathcal{M})$ at time slot t . $d_{u,j}(t) = \|\mathbf{q}_u(t) - \mathbf{h}_j\|_2$, which denotes the spatial distance between UAV u and its nearest threat area $j (j \in \mathcal{J})$ at time slot t .

Similarly to the action space, the observation space is also normalized. The observation data of UAV u at time slot t can be rewritten as

$$o_{u,t} = \{\bar{x}_u(t), \bar{y}_u(t), \bar{z}_u(t), \rho_u(t), \bar{d}_{u,m}(t), \bar{d}_{u,j}(t)\} \quad (34)$$

where $[\bar{x}_u(t), \bar{y}_u(t), \bar{z}_u(t)]$ denotes the normalized UAV position coordinates, which are defined as

$$[\bar{x}_u(t), \bar{y}_u(t), \bar{z}_u(t)] = \left[\frac{x_u(t)}{X^{\text{Max}}}, \frac{y_u(t)}{Y^{\text{Max}}}, \frac{z_u(t)}{Z^{\text{Max}}} \right] \quad (35)$$

$\bar{d}_{u,m}(t)$ is defined as

$$\bar{d}_{u,m}(t) = \frac{d_{u,m}(t)}{\sqrt{(X^{\text{Max}})^2 + (Y^{\text{Max}})^2 + (Z^{\text{Max}})^2}} \quad (36)$$

$\bar{d}_{u,j}(t)$ is defined as

$$\bar{d}_{u,j}(t) = \frac{d_{u,j}(t)}{\sqrt{(X^{\text{Max}})^2 + (Y^{\text{Max}})^2 + (Z^{\text{Max}})^2}} \quad (37)$$

Thus, the observation space $\mathcal{O}$ is defined as

$$\mathcal{O} = \{o_{u,t} | u \in \mathcal{U}, t \in \mathcal{T}\} \quad (38)$$

3.1.3. State space $\mathcal{S}$

The state space contains observations of all agents. Thus, for the state $s_t (s_t \in \mathcal{S})$ at time slot t , it is defined as

$$s_t = \{o_{u,t} | u \in \mathcal{U}\} \quad (39)$$

Thus, the state space $\mathcal{S}$ is defined as

$$\mathcal{S} = \{s_t | t \in \mathcal{T}\} \quad (40)$$

3.1.4. Reward function $\mathcal{R}$

The design of the reward function consists of five parts, which are task rewards, communication rewards, distance rewards, guidance rewards and some penalties.

Task rewards are rewards that the UAV receives for completing the corresponding task, which are defined as

$$R^{\text{task}} = \lambda^{\text{task}} \sum_{u \in \mathcal{U}} \sum_{m \in \mathcal{M}} R_u^{\text{task}}(m) \quad (41)$$

where λ^{task} denotes the weight parameter of task rewards, $R_u^{\text{task}}(m)$ denotes the reward that UAV u receives after participating in the execution of task m , and $R_u^{\text{task}}(m)$ is defined as

$$R_u^{\text{task}}(m) = \begin{cases} \xi_u^{\text{accomplish}}, & \text{if task } m \text{ is executed successfully} \\ \xi_u^{\text{incomplete}}, & \text{otherwise} \end{cases} \quad (42)$$

where $\xi_u^{\text{accomplish}}$ denotes the reward obtained when UAV u participates in the execution of the task m and the task is executed successfully, and

$\xi_u^{\text{incomplete}}$ denotes the penalty obtained when UAV u participates in the execution of the task m but the task failed to execute.

Weight selection and robustness: The overall reward is a weighted sum of multiple components (task, communication, guidance/distance, and penalties). We acknowledge that reward shaping weights can influence learning behavior. In this work, the weights are selected according to two practical principles: (i) scale normalization, where each component is bounded/normalized to a comparable numerical range to prevent any single term from dominating policy gradients; and (ii) behavioral prioritization, where task accomplishment and hard-constraint satisfaction are treated as primary objectives, while communication and guidance terms serve as auxiliary shaping signals. Unless otherwise stated, the same weight settings are used across all compared methods and scenarios for fair comparison.

Communication rewards are rewards obtained by UAVs after successfully accessing communication channels, then the communication reward obtained by UAV u at time slot t is defined as

$$R_u^{\text{access}} = \lambda^{\text{access}} R^{\text{access}}(t) \quad (43)$$

where λ^{access} is the weight parameter of communication rewards, $R^{\text{access}}(t)$ denotes the communication reward obtained by UAVs at time slot t , and $R^{\text{access}}(t)$ is defined as

$$R^{\text{access}}(t) = \begin{cases} \xi^{\text{access}}, & \text{if UAV } u \text{ successfully accesses the communication channel at time slot } t \\ \xi^{\text{fail}}, & \text{otherwise} \end{cases} \quad (44)$$

where ξ^{access} denotes the reward for the UAV successfully accessing the communication channel, and ξ^{fail} denotes the penalty for the UAV not accessing the communication channel.

Distance rewards are flight path length rewards that UAVs receive for selecting their actions, which optimize UAVs' flight trajectories and are defined as

$$R_u^{\text{distance}}(t) = \lambda^{\text{distance}} \|\mathbf{q}_u(t+1) - \mathbf{q}_u(t)\|_2 \quad (45)$$

where $\lambda^{\text{distance}}$ denotes the weight parameter of the distance reward obtained by the UAV while flying.

Guidance rewards can motivate UAVs to fly to task points to perform the corresponding tasks in a shorter time, and also can significantly improve the efficiency of UAVs' tasks execution, which are defined as

$$R_u^{\text{guide}}(t) = \lambda^{\text{guide}} R^{\text{guide}}(t) \quad (46)$$

where λ^{guide} denotes the weight parameter of guidance rewards, $R^{\text{guide}}(t)$ denotes the guidance reward obtained by the UAV at time slot t , and $R^{\text{guide}}(t)$ is defined as

$$R^{\text{guide}}(t) = \begin{cases} \xi^{\text{close}}, & \text{if UAV } u \text{ is close to the target point} \\ \xi^{\text{stay_away}}, & \text{otherwise} \end{cases} \quad (47)$$

where ξ^{close} denotes the reward obtained when the UAV flies in the direction of the target point, and $\xi^{\text{stay_away}}$ denotes the penalty obtained when the UAV hovers in place or flies in a direction away from the target point.

In addition, when the UAV violates predetermined constraints during flight, it needs to be given a certain penalty so as to induce the UAV to rationally adjust its current strategy in order to substitute a more optimal strategy. Therefore, the penalty obtained by UAV u at time slot t is defined as

$$R_u^{\text{penalty}}(t) = \lambda^{\text{safe}} \xi^{\text{safe}} + \lambda^{\text{angle}} \xi^{\text{angle}} + \lambda^{\text{threat}} \xi^{\text{threat}} \quad (48)$$

where λ^{safe} , λ^{angle} and λ^{threat} denote the penalty weight parameters when violating the UAV safety distance, UAV turning angle and threat area constraints respectively, ξ^{safe} , ξ^{angle} and ξ^{threat} denote the penalties in

case of violating the constraints of UAV safety distance, UAV turning angle and threat area respectively, and the penalties of one of them are 0 if constraints of that item are not violated.

It is worth noting that weight parameters are introduced in both the reward and penalty terms. The advantage of designing independent weight parameters lies in the ability to flexibly adjust the priority of corresponding reward terms by modifying these parameters, thereby reducing the complexity of the system in handling optimization problems. Additionally, we combine reward and penalty terms through linear combination to obtain the final reward function. This design approach enhances gradient stability, effectively mitigating convergence difficulties during training.

Therefore, the total reward obtained by UAV u at time slot t is defined as

$$r_u(t) = R_u^{\text{access}}(t) + R_u^{\text{distance}}(t) + R_u^{\text{guide}}(t) - R_u^{\text{penalty}}(t) \quad (49)$$

To reduce sensitivity to absolute magnitudes, each reward component is designed to be bounded (via normalization/clipping) so that their typical values fall into comparable ranges. This design helps stabilize learning and makes the weight parameters interpretable as relative priorities rather than compensating for scale mismatches.

3.1.5. Transfer probability P_r

In our model, the transfer probability is defined as

$$P_r : S \times \mathcal{A} \times S \rightarrow [0, 1] \quad (50)$$

In general, P_r is unknown and it is usually hard to find a specific expression in the learning environment of UAVs, but it can generally be solved by the strategy gradient method.

In each round of generations, the UAV u explores the environment at time slot t and obtains observations $o_{u,t} : S \rightarrow \mathcal{O}_u$ from the environment, where $\mathcal{O}_u \subset S$, and there exists $\rho : S \rightarrow [0, 1]$.

Each UAV selects a corresponding action $a_{u,t}$ based on the current state s_t . The joint action a_t is obtained by information integration through the BS, and the UAV cluster executes the corresponding action to obtain the maximum reward value and finds the optimal policy π_u through continuous generation. Therefore, the multi-UAV cooperative task scheduling and trajectory optimization problem under communication constraints can be represented as

$$\begin{aligned} P2 : \max_{\pi_u} \sum_{u \in U} J(\pi_u) \\ = \max \left\{ \sum_{u=0}^U \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_u(t+k) \right] + R^{\text{task}} \right\} \end{aligned} \quad (51)$$

where γ^k is a discount factor whose main role is to balance current and future rewards.

3.2. TTS algorithm analysis

In order to solve the problem of multi-UAV cooperative task scheduling and trajectory optimization under communication constraints, this paper proposes a reinforcement learning algorithm based on Proximal Policy Optimization (PPO) algorithm for multi-UAV task scheduling, i.e., TTS algorithm.

In the traditional MARL algorithmic framework, an RL framework is often designed for all agent to learn. Although this method has a concise structure and is easy to implement, its main drawback is that each RL learning system is independent of each other and lacks synergy between agents, so unnecessary conflicts may occur between agents during the training process. In this paper, TTS algorithm is designed to solve the problem well, in TTS algorithm, each UAV communicates with the BS or other UAVs so as to obtain the global information within the current time, and then make a decision. Since each UAV has access to the information of the other UAVs before making decisions, all UAVs have

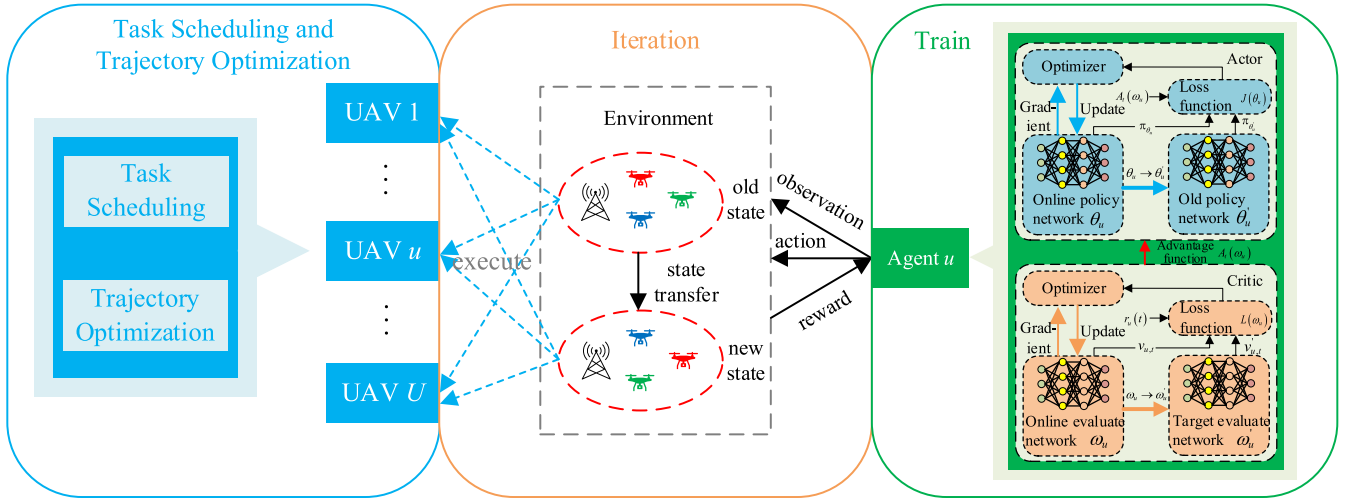


Fig. 2. Schematic of the learning system in multi-UAV cooperative task scheduling and trajectory optimization model under communication constraints.

some synergy in making decisions rather than making decisions independently.

The schematic of the learning system in this chapter is shown in Fig. 2, which integrates the learning networks of all agents into a single MARL framework for learning. During the training process, UAVs obtain current global information by interacting with other UAVs or the BS. In this way, the synergy between UAVs is enhanced. The network structure of TTS algorithm is based on Actor-Critic(AC) architecture, and MLP and GRU are introduced on top of it to achieve more accurate estimation of output results.

The detailed description of the learning system is as follows

3.2.1. AC network

In the traditional AC framework, actions are generated by feeding the current state into Actor network, and then the state and actions are fed into Critic network for value evaluation. This framework tends to be applicable to discrete state space and action space, but for continuous state space and action space, computational inefficiency as well as training instability usually occurs when employing the framework for training of agents. To solve the problem, this paper designs a hybrid Deep Neural Networks (DNN) architecture consisting of MLPs and GRUs, where the role of MLPs are to perform policy function fitting and the role of GRUs are to extract empirical correlations. The learning system in this paper adopts a dual-network structure, which are the online policy (Actor) network and the old policy network, both of which have the same structure but complementary functions; the online policy network is mainly used to carry out the fitting of the strategy function, while the old policy network is mainly used to provide stability guarantees for the strategy, and two networks work together to optimize the decision-making ability of the agent in the continuous space. When a Gaussian distribution is used to describe the strategy, the strategy function of the agent u can be expressed as

$$\pi_{\theta_u}(o_u, h_u^\pi) = \frac{1}{\sqrt{2\pi}\sigma_u(o_u, h_u^\pi)} \cdot \exp\left(-\frac{(a_u - \mu_u(o_u, h_u^\pi))^2}{2(\sigma_u(o_u, h_u^\pi))^2}\right) \quad (52)$$

where θ_u is the online policy network parameter of the agent u , o_u is the observation of the agent u , h_u^π denotes the short-term memory contained in GRU module of Actor network, $\mu_u(o_u, h_u^\pi)$ and $\sigma_u(o_u, h_u^\pi)$ denote the mean and standard deviation of the randomized strategy output by Actor network respectively, and $\mu_u(o_u, h_u^\pi) = \tanh(u)$, $\sigma_u(o_u, h_u^\pi) = \ln(1 + e^u)$, both are activation functions for the output layer of online policy network.

In addition to the online policy network π_{θ} , this paper also introduces the old policy network $\pi_{\theta'}$, whose parameters are updated by periodically copying parameters of the online policy network instead of real-time gradient updating. This mechanism achieves efficient utilization of sample data in the replay buffer by maintaining the difference between the old policy and the current one, which ensures the stability of the policy update and significantly improves the sampling efficiency.

The Critic module also employs a hybrid network architecture consisting of MLPs and GRUs, which realizes the stable evaluation of policies through a dual evaluation (Critic) network, i.e., an online evaluation network and a target evaluation network. The online evaluation network ω_u computes the advantage function regularized by Gaussian noise in real time, and its low variance property effectively overcomes the instability of training process. The target evaluation network ω'_u provides the baseline evaluation value, which is updated by periodically synchronizing parameters of the online evaluation network. The two networks work together to suppress the tendency of traditional advantage functions to overfit, and also achieve smooth optimization of the advantage estimation by incorporating a random noise mechanism, thus providing more reliable feedback of the action value to the Actor module. As a result, the output of the online evaluation network of the agent u is defined as

$$v_{u,t} = V_{\omega_u}(\tilde{s}_t, h_u^V) \quad (53)$$

where V_{ω_u} denotes the value function of the online evaluation network of the agent u and ω_u denotes the parameters of the online evaluation network of the agent u , $\tilde{s}_t = \text{concat}(s_t, \tilde{n})$, $\text{concat}(\cdot, \cdot)$ is the connectivity function, $\tilde{n} \sim \mathcal{N}(0, 1)$ denotes Gaussian noise, h_u^V denotes short-term memories contained in the GRU module. Therefore, the value function of the target evaluation network of the agent u is $V_{\omega'_u}$ and its output is $v'_{u,t}$, the target evaluation network shares network parameters with the online evaluation network.

In this paper, we used Generalized Advantage Estimation (GAE) method to estimate the advantage function, and GAE method achieves a dynamic balance between variance and bias. Thus, the dominance function in the Critic module of the agent u is defined as

$$A_{u,t} = \sum_{k=t}^{T-1} (\gamma \lambda^{\text{GAE}})^{k-t} \zeta_{u,k} \quad (54)$$

where γ denotes the discount factor, λ^{GAE} is the attenuation factor for GAE, which is used to adjust the variance and bias of the return estimates. $\zeta_{u,k}$ is defined as

$$\zeta_{u,t} = r_t + \gamma v'_{u,t+1} - v'_{u,t} \quad (55)$$

where r_t denotes the total reward value of all agents at time slot t , i.e.

$$r_t = \sum_{u \in U'} r_{u,t} \quad (56)$$

3.2.2. Explore enhancement methods

Adequate environment exploration is crucial in MARL learning systems, and traditional gradient-based approaches often lead to insufficient exploration. Although many solutions exist, including greedy algorithms, adding noise to action vectors to encourage agents to explore the environment and other methods that can effectively enhance the exploration ability of value function or deterministic policy-based algorithms, they are not applicable to the learning system that outputs stochastic policies in this paper.

In this paper, firstly, the exploration of the environment by agents is enhanced by designing a reasonable reward function, and then the exploration of the environment by agents is encouraged by introducing an entropy loss function in the stochastic policy, and the combination of two methods enable agents to explore the learning environment more fully. In TTS algorithm, the entropy loss function of agent u is defined as

$$f^E(\pi_{\theta_u}) = \mathcal{H}(\pi_{\theta_u}) \quad (57)$$

where $\mathcal{H}$ is the entropy function. The introduction of policy entropy can motivate agents to learn different actions, not limited to certain actions, as a way to enhance agents' exploration of the environment.

3.2.3. Training process

The learning system proposed in this paper is trained end-to-end using AC framework, where the online policy network performs parameter updating through the policy gradient method, while the online evaluation network optimizes value estimation through Temporal Difference Learning (TDL). Therefore, according to the definition of the online policy network parameters π_{θ_u} and the advantage function, the gradient of the online policy network is defined as

$$\nabla_{\theta} J(\theta) = \sum_{u \in U'} \mathbb{E}_t \left[\nabla_{\theta_u} (\log \pi_{\theta_u}) A_{u,t} \right] \quad (58)$$

where $\mathbb{E}_t[\cdot]$ is the sampling average.

In addition, for the shortcomings of traditional policy gradient methods with large variance and slow convergence speed, this paper constructs the ratio weights of the old and new policies through the importance sampling technique, and makes full use of historical samples to improve the sampling efficiency during policy updating, and also strictly restricts the policy updating amplitude by a clipping mechanism of the advantage function. This clipping mechanism avoids certain factors that may cause instability during policy updates by limiting the maximum step size of policy updates, these factors include over-optimization and over-penalization. Preventing over-optimization avoids the policy becoming overly greedy by favoring better actions, thereby reducing necessary exploration. Preventing over-penalization avoids the policy completely forgetting worse actions, reducing the possibility of re-evaluating them in the future. Consequently, introducing the clipping mechanism significantly improves the algorithm's stability. Therefore, the loss function of the online policy network for agent u is defined as

$$\begin{aligned} L^A(\theta_u) = & \frac{1}{N^B} \sum_t^{t+N^B} \left[\min(\mathcal{X}(\theta_u) A_{u,t}, \right. \\ & \left. f^{CL}(\mathcal{X}(\theta_u), 1 - \epsilon, 1 + \epsilon) A_{u,t}) \right] \\ & + \beta \frac{1}{N^B} \sum_t^{t+N^B} f^E(\pi_{\theta_u}) \end{aligned} \quad (59)$$

where N^B is the batch size of the sampled data, $\mathcal{X}(\theta_u)$ is the ratio of the probability of the new strategy to the old one, f^{CL} is the truncation

function that restricts $\mathcal{X}(\theta_u)$ to between $[1 - \epsilon, 1 + \epsilon]$, ϵ is the truncation parameter and β is the weight parameter. $\mathcal{X}(\theta_u)$ is defined as

$$\mathcal{X}(\theta_u) = \frac{\pi_{\theta_u}}{\pi_{\theta_u'}} \quad (60)$$

The loss function of the online evaluation network for the agent u is defined as

$$L^C(\omega_u) = \frac{1}{N^B} \sum_t^{t+N^B} \left(y_{\omega_u'}(\tilde{s}_t) - V_{\omega_u}(\tilde{s}_t) \right)^2 \quad (61)$$

where $y_{\omega_u'}(\tilde{s}_t)$ is defined as

$$y_{\omega_u'}(\tilde{s}_t) = A_{u,t} + V_{\omega_u'}(\tilde{s}_t) \quad (62)$$

The way in which parameters of the online policy network and the online evaluation network are updated is defined as follows

$$\theta_u = \theta_u + \alpha^A \nabla_{\theta_u} L^A(\theta_u) \quad (63)$$

$$\omega_u = \omega_u - \alpha^C \nabla_{\omega_u} L^C(\omega_u) \quad (64)$$

Where α^A and α^C are the learning rate parameters of the online policy network and online evaluation network respectively, which determine the convergence rate of TTS algorithm.

3.2.4. Testing process

In the testing phase, the trained online policy network will be applied to make decisions about UAVs without relying on the online evaluation network for assisted decision making. In the test environment, each UAV is first loaded with parameters of the trained and converged Actor network, and then based on the observation information of the real-time environment, successive actions are generated by the Actor network, and finally by executing the corresponding actions in order to accomplish tasks in the environment.

3.2.5. Complexity analysis

This subsection provides a specific analysis of the computational complexity of TTS algorithm, which is mentioned in the multi-UAV collaborative task scheduling and trajectory optimization system under communication constraints. The complexity of TTS algorithm is divided into two parts, model training complexity and model testing complexity.

In TTS algorithm, the computational complexity of the model training part is $\mathcal{O}(\frac{E \cdot T}{B \cdot L \cdot N})$, where E is the maximum number of iterations, T is the maximum number of training steps, B is the sampling batch size, L denotes the time complexity of each layer, and N denotes the number of layers. And the computational complexity of the model testing part of TTS algorithm can be approximated as $(\mathcal{O}^M(m) + \mathcal{O}^A(m))N$, where $\mathcal{O}^M(m)$ denotes the computational complexity of the matrix operation for each layer with a node number of m , $\mathcal{O}^A(m)$ denotes the computational complexity of the activation function for each layer with a node number of m , and N also denotes the number of layers. Therefore, based on the above analysis, the total computational complexity of TTS algorithm proposed in this paper is approximated as $\mathcal{O}(\frac{E \cdot T}{B \cdot L \cdot N}) + (\mathcal{O}^M(m) + \mathcal{O}^A(m))N$.

The details of TTS algorithm are shown in Algorithm 1. In this paper, TTS algorithm is employed to solve the problem of multi-UAV cooperative task scheduling and trajectory optimization under communication constraints.

The entire TTS algorithm consists of two parts: the data collection component (lines 2–15) and the network update component (lines 16–29). Each UAV selects action $a_{u,t}$ and executes it, obtaining the next moment's local observation data $o_{u,t+1}$. After all UAVs complete their actions and acquire local observation data, information is integrated via the BS to compute the reward value r_t . It should be noted that if a particular UAV u violates the aforementioned flight constraints, the data for UAV u must be deleted from the current record, and UAV u revert to the previous state and execute a new action (lines 8–10). After all data collection has been completed, the network update operation will

be performed. For each UAV's collected data, a fixed batch of data is sampled for network updates. The parameter of online policy network θ_u and the parameter of online evaluation network ω_u are updated by calculating their loss functions, then update the parameter of old policy network θ'_u and the parameter of target evaluation network ω'_u (lines 21–27).

Algorithm 1 Temporal task scheduling algorithm.

Initialize: Discount factor γ , truncation parameter ϵ , Actor network learning rate α^A , Critic network learning rate α^C , GAE decay factor λ^{GAE} , reward factor λ^{task} , λ^{access} , $\lambda^{\text{distance}}$ and λ^{guide} , penalty factor λ^{safe} , λ^{angle} and λ^{threat} , reward function $R^{\text{access}}(t)$ and $R^{\text{guide}}(t)$, maximum number of iterations N^{max} , maximum time steps T , total number of UAVs U , sampling batch size N^B , maximum number of epochs N^E

Initialize: Actor network parameter θ and θ' , Critic network parameter ω and ω'

Output: The optimal sequence of actions associated with the flight trajectory of each UAV

```

1: for episode  $e \leftarrow 1$  to  $N^{\text{max}}$  do
2:   Initialize the environment, obtain the initial state  $s_0$ , and initialize
   the communication index  $c_{u,t}$  for each UAV
3:   for step  $t \leftarrow 1$  to  $T$  do
4:     for UAV  $u \leftarrow 1$  to  $U$  do
5:       Select action  $a_{u,t}$ 
6:       Execute action  $a_{u,t}$ , record  $c_{u,t}$ , get next observation  $o_{u,t+1}$ 
7:       if The action of UAV  $u$  violates constraints then
8:         Cancel this move by UAV  $u$ 
9:         Calculate penalties  $R_u^{\text{penalty}}(t)$ 
10:        Execute rollback operation  $o_{u,t+1} \leftarrow o_{u,t}$ , cancel record  $c_{u,t}$ 
11:      end if
12:       $o_{u,t} \leftarrow o_{u,t+1}$ 
13:    end for
14:    Calculate connection state  $\tilde{s}_t$  and reward  $r_t$ 
15:  end for
16:  for UAV  $u \leftarrow 1$  to  $U$  do
17:    Obtain sampled data sets  $\{o_{u,t}, a_{u,t}, \tilde{s}_t, r_t\}_{t=1}^T$ 
18:    Estimate the advantage function  $A_{u,t}$  in the Critic module
19:    for all batch  $j \leftarrow 1$  to  $\frac{T}{N^B}$  do
20:      Select the  $j$ th batch of sampled data sets  $\{o_{u,t}, a_{u,t}, \tilde{s}_t, r_t\}_{t=j}$ 
21:      for epoch  $k \leftarrow 1$  to  $N^E$  do
22:        Calculate the loss function  $L^A(\theta_u)$  of the online policy network
23:        Calculate the loss function  $L^C(\omega_u)$  of the online evaluation network
24:        Optimize online evaluation network parameters  $\omega_u$ 
25:        Optimize online policy network parameters  $\theta_u$ 
26:      end for
27:      Update parameters of old policy network and target evaluation network:  $\theta'_u \leftarrow \theta_u$ ,  $\omega'_u \leftarrow \omega_u$ 
28:    end for
29:  end for
30: end for

```

TTS adopts PPO-style clipped policy optimization with GAE and entropy regularization. The key difference lies in a constraint-aware on-policy data collection mechanism tailored to our hard-constrained UAV setting: whenever an action violates any hard flight constraint (e.g., safety distance, turning-angle limits, threat avoidance, or time-window feasibility), the action is cancelled, the agent rolls back to the previous feasible state, and the corresponding transition is removed from the on-policy record before re-sampling a new action. This rollback and record cancellation procedure changes the effective sampling distribution and prevents infeasible samples from entering policy updates, which is critical for stable learning under multiple simultaneous hard constraints. In addition, we implement a BS-assisted information integration proto-

col consistent with the one-hop communication assumption, where the joint action is constructed based on aggregated global information, reflecting the heterogeneous information availability between in-coverage and out-of-coverage UAVs.

4. Simulation analysis

In this section, we established a simulation scenario for multi-UAV cooperative task scheduling and trajectory optimization under communication constraints, and verified the effectiveness and the stability of TTS algorithm proposed in this paper.

We note that the current evaluation primarily focuses on learning convergence and task-level performance (e.g., task completion rate and trajectory length). Communication-related metrics such as explicit communication overhead, coordination delay, or scheduling latency are not quantitatively reported in this manuscript. We acknowledge this as a limitation, and we will include such communication-centric metrics in future work to more directly support the communication-constrained motivation.

4.1. Parameter setting

In the simulation experiment conducted in this paper, we designed two distinct simulation scenarios, and restrict all UAVs to fly within a three-dimensional space of size $500 \times 500 \times 200\text{m}^3$, with a task cycle set to $T = 100$ and the coordinates of the BS location set to $[250, 250, 0]$. The location of the task target point appears randomly in the scenario, and its appearance time is also random, and UAVs need to optimize their flight trajectories and complete as many tasks as possible.

The parameter settings involved in the simulation experiment described in this paper are as follows. Safety distance between UAVs $d^S = 5\text{m}$, maximum turning angle $\theta = 45^\circ$, task area radius $R^M = 10\text{m}$, the BS signal coverage radius $R^B = 200\text{m}$, the length of each time slot $\Delta\tau = 1\text{s}$, spectrum sensing duration $\tau = 0.1\text{s}$, signal transmission power of the BS $P_B = 300\text{mW}$, signal transmission power of UAVs $P_U = 20\text{mW}$, path loss index $\alpha = 3$, attenuation factor $\eta = 0.1$. Task completion reward $\xi_m^{\text{accomplish}} = 10$, penalty for uncompleted tasks $\xi_m^{\text{incomplete}} = -3$, reward for successfully connecting UAVs to communication channels $\xi^{\text{access}} = 3$, penalty for UAVs not connected to communication channels $\xi^{\text{fail}} = -1$, reward for UAVs approaching target points $\xi^{\text{close}} = 1.5$, penalty for UAVs not moving or moving away from target points $\xi^{\text{stay_away}} = -1$, reward factor $[\lambda^{\text{access}}, \lambda^{\text{distance}}, \lambda^{\text{guide}}] = [0.3, 0.1, 0.6]$, penalty factor $[\lambda^{\text{safe}}, \lambda^{\text{angle}}, \lambda^{\text{threat}}] = [0.4, 0.25, 0.35]$. Discount factor $\gamma = 0.95$, clip parameter $\epsilon = 0.2$, GAE attenuation factor $\lambda^{\text{GAE}} = 0.95$, Actor network learning rate $\alpha^A = 0.001$, Critic network learning rate $\alpha^C = 0.001$. The neural network has 2 hidden layers, each containing 64 neurons. The GRU step size is set to 100, and the batch size for sampling is 256.

4.2. Convergence analysis

In this subsection, we introduce three comparison algorithms: the TTS-KL algorithm, the RTTS algorithm, and the classical Multi-Agent Proximal Policy Optimization (MAPPO) algorithm. We then compare the TTS algorithm with these three comparison algorithms to validate the effectiveness and reliability of the TTS algorithm proposed in this paper. Brief introductions to the TTS-KL algorithm and the RTTS algorithm are as follows:

TTS-KL: The KL divergence method is employed to update the neural network, which is a multi-agent algorithm.

RTTS: The random algorithm is employed for action selection, which is a multi-agent algorithm.

Fig. 3(a) and (b) show the convergence performance comparison results between TTS algorithm and the three aforementioned comparison algorithms. Among them, Fig. 3(a) shows the comparison results of the reward curves. Compared with TTS-KL algorithm and MAPPO algorithm, TTS algorithm converges faster and achieves a higher cumulative

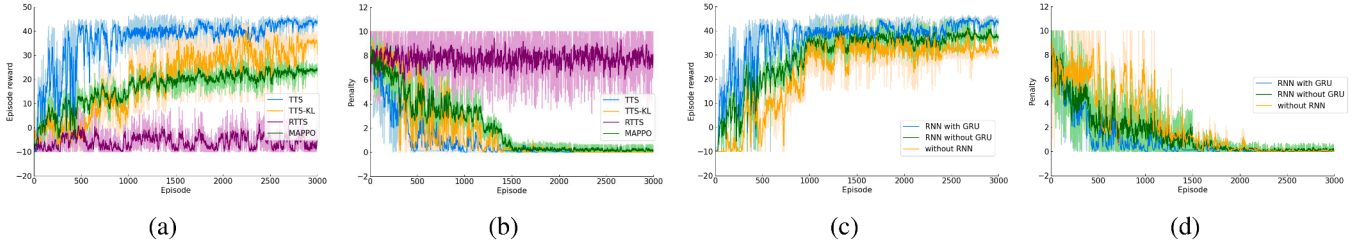


Fig. 3. Comparison of algorithm convergence performance.

reward value during training. Additionally, the reward curves for all three algorithms exhibit significant volatility in the early stages. As the number of iterations increases, the fluctuations in the TTS algorithm's reward curve gradually decrease, while reward curves of TTS-KL algorithm and MAPPO algorithm continue to exhibit some fluctuations even after convergence. Compared with RTTS algorithm, the overall performance of TTS algorithm is more stable, while the reward curve of RTTS algorithm shows significant fluctuations throughout the training process. It is clear from the figure that RTTS algorithm still cannot converge to a high reward value at the end of training. Fig. 3(b) shows the comparison results of penalty curves. As can be seen from the figure, TTS algorithm still demonstrates superior convergence performance.

It is worth noting that although TTS-KL algorithm and MAPPO algorithm ultimately converge to lower penalty values, during the early to mid stages of training, the penalty values obtained by these algorithms consistently exceed those of the TTS algorithm. RTTS algorithm performs extremely unstable throughout the training process, with consistently high penalty values, and ultimately fails to converge. Overall, throughout the entire training process, TTS algorithm significantly outperforms the other three comparison algorithms in terms of acquired rewards and penalties, and it also demonstrates superior convergence performance. Although TTS-KL algorithm and MAPPO algorithm performed reasonably well, they obtained lower cumulative reward values than TTS algorithm during training and converged more slowly than TTS algorithm. RTTS algorithm, on the other hand, fails to converge throughout the training process and receives more punitive negative rewards, indicating its overall poor performance.

Fig. 3(c) and (d) show the convergence performance comparison of TTS algorithm with and without the introduction of RNN and GRU. Fig. 3(c) shows the comparison result of reward curves, as can be seen from the figure, compared to scenarios where RNN is introduced without GRU and where neither RNN nor GRU are introduced, TTS algorithm that incorporates both RNN and GRU demonstrates superior performance in convergence speed and episode reward values. After convergence, the overall performance of TTS algorithm with RNN and GRU is more stable, and the reward curve fluctuates less. Fig. 3(d) shows the comparison result of penalty curves. Compared to the two situations mentioned above, the penalty value obtained by TTS algorithm with RNN and GRU shows a significant downward trend in the early stages of training, and after a certain number of iterations, the fluctuations in its penalty curve are significantly reduced. In contrast, TTS algorithm with RNN but without GRU, as well as TTS algorithm without RNN, require more iterations to converge, and they receive more punitive negative rewards during training, resulting in poorer stability. Therefore, integrating memory by introducing RNN and GRU can significantly improve the training efficiency, stability, and overall performance of TTS algorithm.

In addition, since the discount factor γ is a very important parameter in RL algorithms, it is used to measure the importance of future rewards when the agent learns the current strategy. When γ is small, the agent will consider rewards obtained in the present more in the learning process, and ignore the impact of future rewards on the current strategy to a certain extent, causing the agent to become short-sighted and its training effect to deteriorate. Conversely, when γ is large, the agent

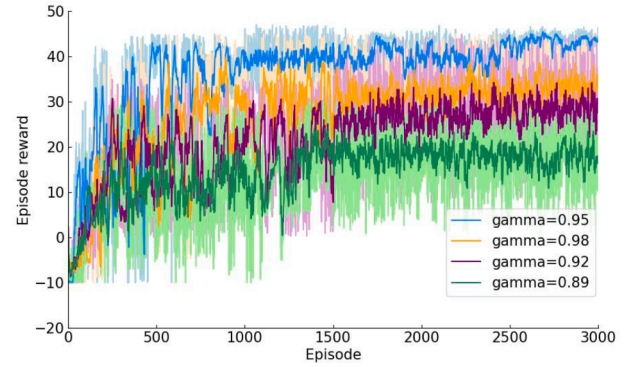


Fig. 4. Comparison results of convergence performance of TTS algorithm with different sizes of discount factor γ .

places greater emphasis on future rewards during learning, while the influence of current rewards becomes relatively smaller. This leads to the agent considering too many time steps, making it difficult for the algorithm to converge. Therefore, the appropriateness of the value of γ has a significant impact on the overall performance of RL algorithms. In this subsection, we analyze the convergence of TTS algorithms with different discount factors.

Fig. 4 shows the convergence performance comparison results of TTS algorithm under different discount factors. As shown in the figure, when $\gamma < 0.95$, the episode reward obtained by agents during training increases gradually with the increase in γ , and the convergence speed of TTS algorithm also gradually increases; when $\gamma = 0.95$, the episode reward value obtained by agents during training reaches its maximum, and the convergence speed of TTS algorithm is the fastest; when $\gamma > 0.95$, the overall performance of TTS algorithm begins to deteriorate, the episode reward obtained by agents during training gradually decreases, and the convergence speed of TTS algorithm also decreases to a certain extent. Based on the above analysis, when the discount factor is set to $\gamma = 0.95$, TTS algorithm performs best overall. Therefore, in this paper, we set the discount factor to 0.95.

4.3. Flight trajectory

In this subsection, we designed two distinct simulation scenarios: Scenario I features 3 UAVs and 6 tasks, while Scenario II involves 6 UAVs and 18 tasks. Figs. 5 and 7 respectively illustrate the UAV flight trajectories and communication links obtained using the TTS algorithm in Scenario I and Scenario II. Figs. 6 and 8 respectively show the correspondence between tasks and time slots in Scenario I and Scenario II.

In UAV flight trajectory sequence diagrams, colored cones represent the BS, transparent hemispheres centered on the BS represent the signal coverage range of the BS, and green hemispheres represent threat areas. Different colored lines correspond to different UAV flight trajectories, and all UAVs depart from the BS location. Different colored balls correspond to different types of task locations, which are all labeled in the diagram. In communication link diagrams, colored cones also represent

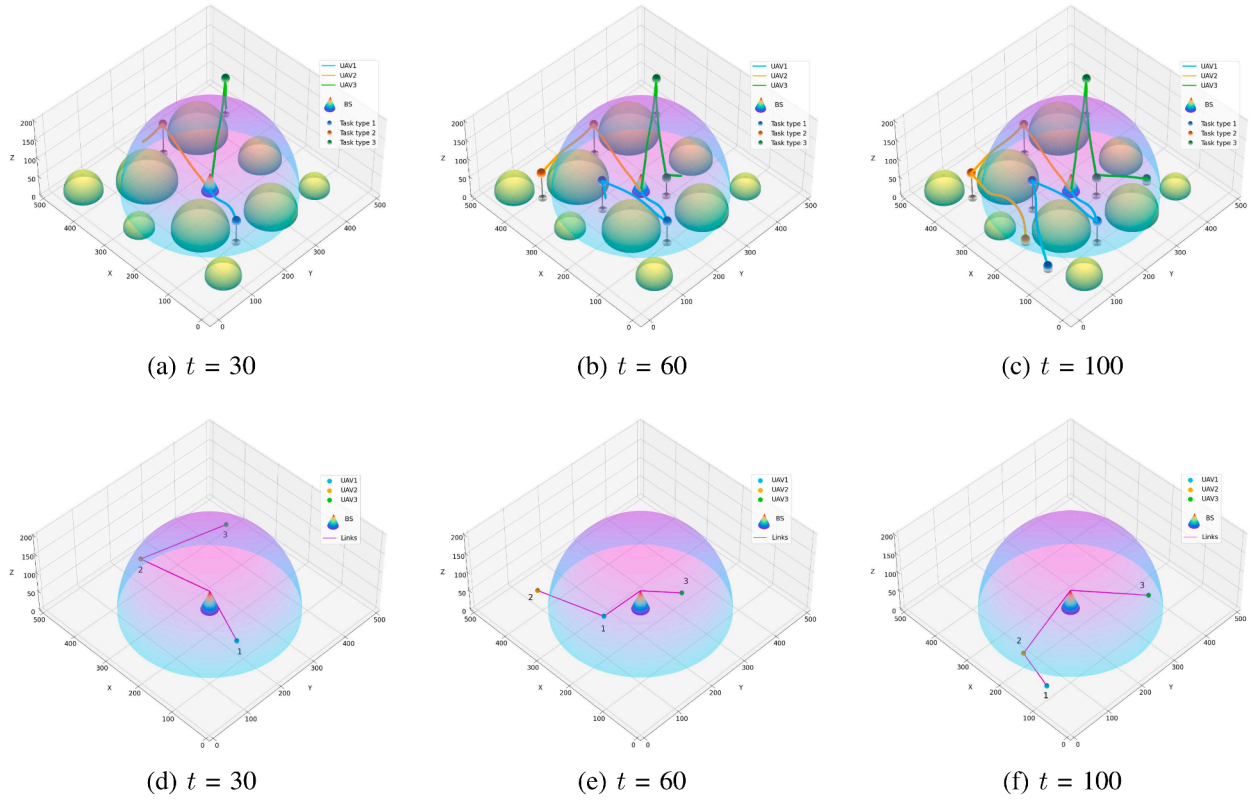


Fig. 5. Flight trajectory diagrams and communication link diagrams of UAVs trained using TTS algorithms in scenario I.

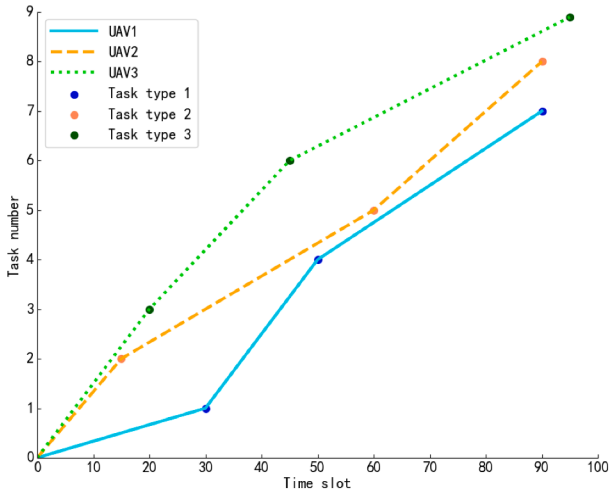


Fig. 6. Relationship diagram between tasks and time slots in scenario I.

the BS, and transparent hemispheres centered on the BS also represent the signal coverage range of the BS. The difference is that small balls of different colors represent the locations of different UAVs, and their coordinates are marked in the diagrams, while pink lines represent two-way communication links.

Referring to Fig. 6, Fig. 5(a) shows that at $t = 30$, UAV 1, UAV 2 and UAV 3 have completed task 1, task 2 and task 3 respectively. At this point, UAV 1 remains near the location of task 1, while UAV 2 and UAV 3 continue flying forward. As shown in Fig. 5(d), both UAV 1 and UAV 2 are within the BS signal coverage area and can communicate directly with the BS. However, UAV 3 is outside the BS signal coverage area and cannot communicate with the BS, after performing spectrum sensing, it successfully accesses the communication channel and can communicate with UAV 2. As shown in Fig. 5(b), at $t = 60$, UAV 2 flies to the vicinity

of task 5's location and completes the task, while UAV 1 and UAV 3 continue forward flight after completing task 4 and task 6 respectively. Fig. 5(e) shows that UAV 1 remains within the BS signal coverage area, while UAV 2 has flown out of the coverage area, UAV 3 has flown from outside the coverage area back into it. Both UAV 1 and UAV 3 can now communicate with the BS, while UAV 2 can only communicate with UAV 1. As shown in Fig. 5(c), at $t = 100$, UAV 1, UAV 2 and UAV 3 completed task 7, task 8 and task 9 respectively, successfully reaching the vicinity of the ground target point. As shown in Fig. 5(f), at this point, UAV 2 and UAV 3 are within the BS signal coverage area and can both communicate with the BS, but UAV 1 is located outside the BS signal coverage area and can only communicate with UAV 2.

Combining Fig. 8, it can be seen from Fig. 7(a) that when $t = 30$, UAVs 1–6 have completed tasks 1–6 respectively. The corresponding communication link diagram is shown in Fig. 7(d). At this point, all UAVs are within the BS signal coverage range and can communicate with the BS. As can be seen from Fig. 7(b), when $t = 60$, UAVs 1–6 completed tasks 7–12 respectively. It can be seen from Fig. 7(e), UAV 2, UAV 4 and UAV 6 are still within the BS signal coverage range, and they can communicate with the BS. However, UAV 1, UAV 3 and UAV 5 have flown outside the range and need to perform spectrum sensing and access the communication channel before they can communicate. UAV 3 communicates with UAV 2, while UAV 1 and UAV 5 communicate with UAV 6. As can be seen from Fig. 7(c), when $t = 100$, UAVs 1–6 completed tasks 13–18 respectively, and flew to the ground target points. It can be seen from Fig. 7(f), at this point, UAV 3 and UAV 6 are within the BS signal coverage range and can communicate with the BS. All other UAVs are outside the range, UAV 2 and UAV 4 communicate with UAV 3, while UAV 1 and UAV 5 communicate with UAV 6.

Additionally, it is worth noting that, for UAV 3 in Fig. 5(c) and UAV 3 and 6 in Fig. 7(c), they both first flew to a task target point farther away from them and then returned while performing the task. This indicates that UAVs do not always choose the task target point closest to them as their destination, but instead opt for a detour flight route. This flight mode effectively proves that the occurrence of tasks is sequential.

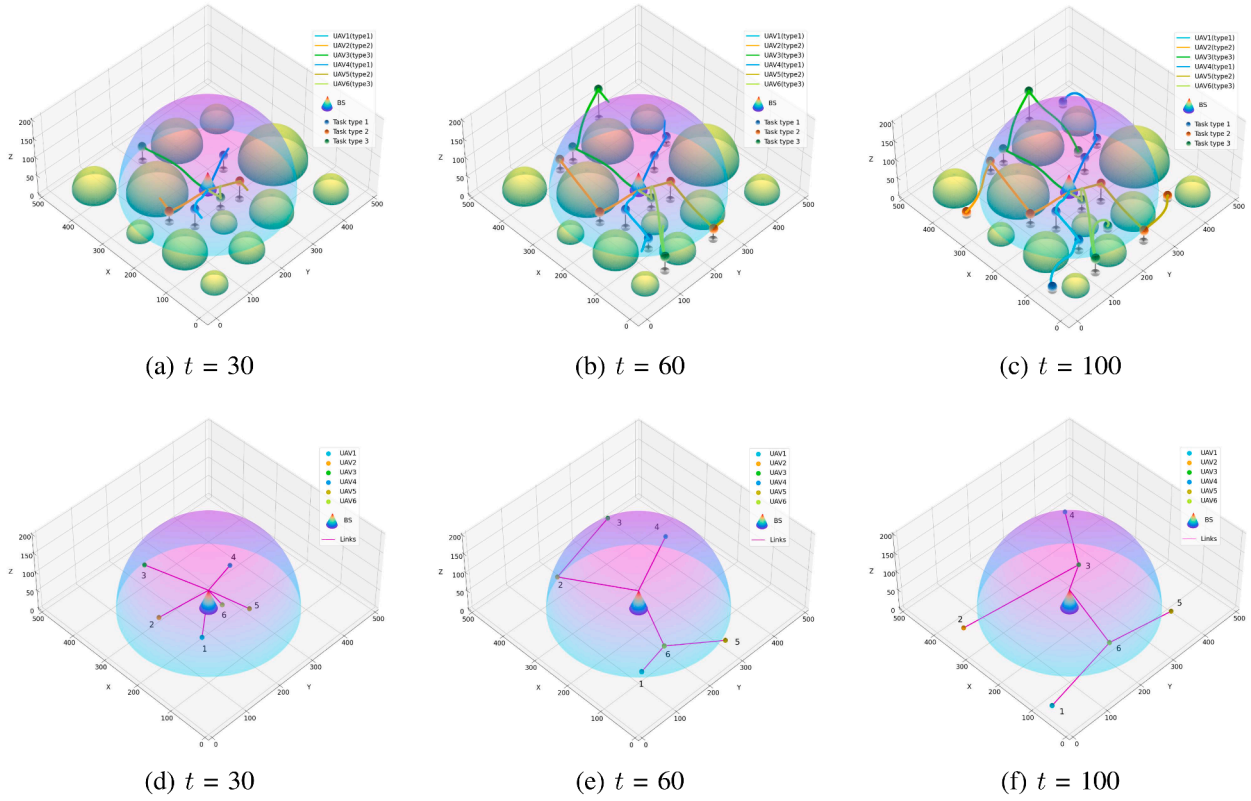


Fig. 7. Flight trajectory diagrams and communication link diagrams of UAVs trained using TTS algorithms in scenario II.

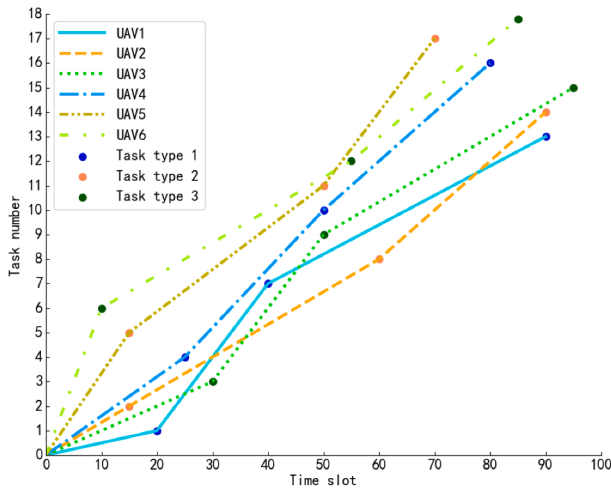


Fig. 8. Relationship diagram between tasks and time slots in scenario II.

The simulation results in the two different scenarios above indicate that TTS algorithm can effectively solve the optimization problem proposed in this paper, namely, multi-UAV cooperative task scheduling and trajectory optimization under communication constraints. While optimizing the flight trajectories of UAVs, it also ensures the communication efficiency between endpoints, thereby ensuring that UAVs can efficiently complete their tasks.

4.4. Comparative analysis

In this subsection, we compare the training times of TTS algorithm, TTS-KL algorithm, RTTS algorithm and MAPPO algorithm under different numbers of UAVs. Additionally, we have also set up three sets of

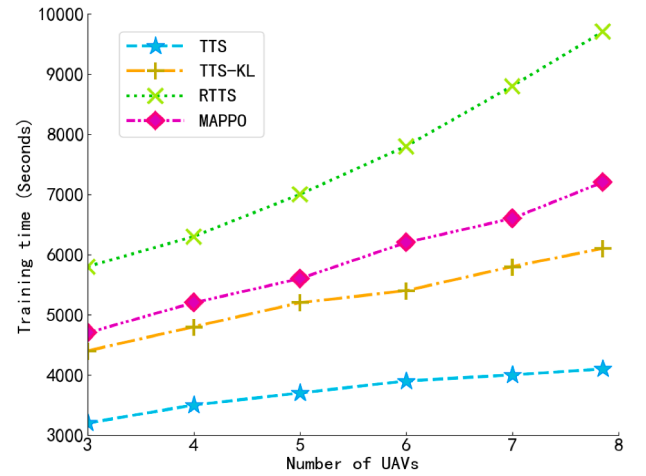


Fig. 9. Comparison diagram of algorithm training times for different numbers of UAVs.

comparative experiments for two important indicators: task completion rate and UAV flight trajectory length, and separately when the BS signal coverage radius, UAV communication range radius and number of UAVs are different, the overall performance of four algorithms mentioned above was compared.

As shown in Fig. 9, the training time for all four algorithms increases to varying degrees as the number of UAVs grows. TTS algorithm consistently requires the least training time, followed by TTS-KL and MAPPO algorithms, while RTTS algorithm requires the most training time. Additionally, when the number of UAVs increases, the training time of TTS algorithm shows a smaller increase compared to the other three comparison algorithms, whose training time increases are relatively larger than that of TTS algorithm. This demonstrates that TTS algorithm

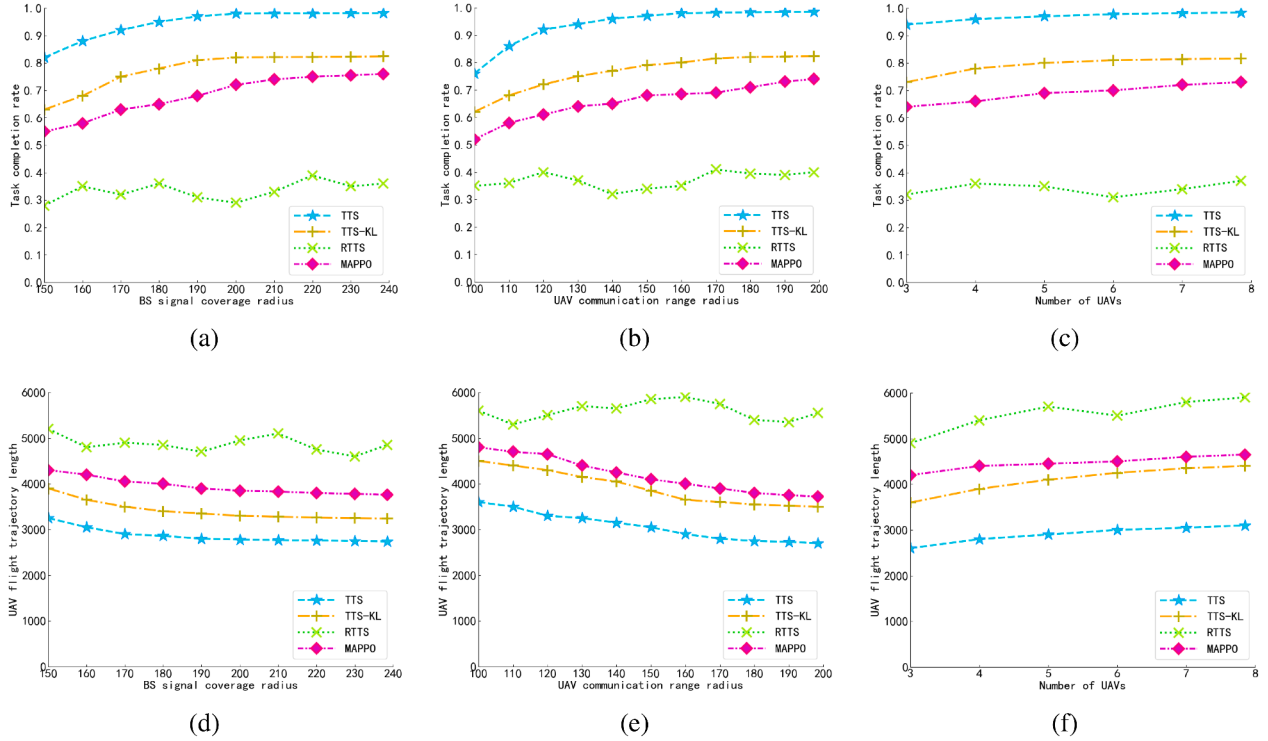


Fig. 10. The results of three sets of comparative experiments were set for two important indicators: task completion rate and UAV flight trajectory length.

exhibits superior training efficiency compared to the aforementioned three comparison algorithms.

As shown in Fig. 10(a), (b) and (c), when the BS signal coverage range, the UAV communication range radius and the number of UAVs increase, the task completion rates of TTS algorithm, TTS-KL algorithm and MAPPO algorithm show an overall upward trend and eventually stabilize. The task completion rate of TTS algorithm is higher than TTS-KL algorithm and MAPPO algorithm, indicating that the overall performance of TTS algorithm is better. The task completion rate curve of RTTS algorithm shows an upward and downward fluctuation trend, and the task completion rate remains at a relatively low level, indicating that the performance of RTTS algorithm is poor.

As can be seen from Fig. 10(d) and (e), when the BS signal coverage range and the UAV communication range radius increase, the flight trajectory lengths of TTS algorithm, TTS-KL algorithm and MAPPO algorithm gradually decrease and eventually stabilize. However, the flight trajectory length obtained using TTS algorithm remains consistently lower than TTS-KL algorithm and MAPPO algorithm. For RTTS algorithm, the UAV flight trajectory length curve remains in a fluctuating state, and its values are consistently high, indicating that RTTS algorithm has poor stability. As shown in Fig. 10(f), when the number of UAVs increases, the flight trajectory length curves of four algorithms all show an upward trend. Compared with the other three comparison algorithms, the flight trajectory length obtained using TTS algorithm is always the shortest, indicating that TTS algorithm has better performance.

Based on the above comparative analysis results, it is evident that the overall performance of TTS algorithm is more ideal than TTS-KL algorithm, RTTS algorithm and MAPPO algorithm. TTS-KL algorithm and MAPPO algorithm performs well in solving multi-UAV cooperative task scheduling problems, achieving high task completion rate and low UAV flight trajectory length. However, it still falls short of TTS algorithm in comparison. Since RTTS algorithm makes random selections when choosing actions, its stability is poor, and it may even fail to converge during training. RTTS algorithm also performs worse overall in the aforementioned comparative experiments. In summary, TTS algo-

rithm performs best in terms of task completion rate and UAV flight trajectory, followed by TTS-KL algorithm and MAPPO algorithm, while RTTS algorithm performs the worst.

5. Conclusion

In this paper, we investigate multi-UAV cooperative task scheduling and trajectory optimization methods under communication constraints. To address this problem, we established a corresponding system model and considered constraints such as UAV performance constraints, task constraints, communication constraints and spectrum sensing constraints to formulate the optimization problem. In terms of algorithm selection, we designed a TTS algorithm based on MARL algorithm framework to solve the optimization problem proposed in this paper. We provided a detailed introduction to the algorithm's design process and specific architectural components, and conducted a complexity analysis of the algorithm. Finally, simulation experiments were conducted, and use the TTS-KL algorithm, RTTS algorithm and MAPPO algorithm as comparison algorithms. Through comparative analysis, the effectiveness of TTS algorithm in addressing multi-UAV cooperative task scheduling and trajectory optimization under communication constraints was validated. This study has several limitations, such as the relatively idealized model established and the limited number of key factors affecting algorithm performance that were incorporated into the analysis. In particular, we did not integrate the GRU-MLP architecture into other representative MARL baselines for broader generalizability validation; this will be addressed in future work. As UAV technology continues to advance, future application scenarios for multi-UAV cooperative task scheduling will grow increasingly complex, posing significant challenges to the generalization capabilities of established models. Therefore, future research should focus on developing system models that more closely align with real-world environments while incorporating additional factors affecting system performance, this enables the design of more efficient algorithms for multi-UAV cooperative task scheduling and trajectory optimization.

CRedit authorship contribution statement

Kehao Wang: Writing – review & editing, Supervision, Conceptualization; **Zhe Cheng:** Writing – original draft, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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